

EDUCATION

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B.A.Sc. in Engineering Science (Robotics) — Professional Experience Year (PEY)

Sep 2022 – Apr 2027

University of Toronto, Toronto, ON

- **Interests:** Transportation infrastructure, rail & transit, project delivery, GIS/spatial analysis.
- **Thesis:** AI-Assisted Conversational Travel Survey for Transportation Data Collection — developing a web-based conversational survey system for structured travel-data collection and response validation.
- **Selected Coursework:** Transportation Planning; Economic Analysis & Decision Making (A+); Embedded Systems (A); Data Structures & Analysis; Control Systems; Computer Vision for Robotics.
- **Tools:** Pavement ME, ArcGIS, Azure Blob Storage, Power Automate, SharePoint, Git, MATLAB.
- **Programming:** Python (pandas, NumPy, scikit-learn), SQL (MySQL), C/C++.

EXPERIENCE

Engineering Associate (PEY Co-op) — Pavement Evaluation & Asset Management

May 2025 – Sep 2026

Ministry of Transportation, North York, ON

- Developed Power Automate workflows for pavement distress data collection and ARAN field-data reporting, automating Microsoft Forms submissions, PDF report generation, SharePoint storage, and management distribution; reduced reporting turnaround **from days to near real time** and saved **8–10 hours/week** of manual work.
- Evaluated five GNSS/INS vendors for a **~\$25,000** system, comparing pricing and technical specifications; authored an MTO procurement request in accordance with internal and Buy Ontario requirements, which received management approval.
- Supported Pavement ME 3.0 local calibration by selecting calibration sections, preparing 2012–2025 MTO pavement condition data and model projects, and validating distress predictions; sensitivity analysis identified annual average daily truck traffic (AADTT) as the most influential factor for fatigue cracking, layer thickness for rutting, and temperature for thermal cracking.
- Modelled the friction number (FN)–speed relationship using ordinary least squares (OLS) regression on 2022 and 2024 MTO friction data, including measurements up to 110 km/h. Co-authored an MTO Technical Report on speed recommendations for 400-series highways.
- Processed and integrated **579,086** Ontario Road Network (ORN) records using Python (pandas) and ArcGIS, joining regional datasets by unique road-segment identifiers to add road names, speed limits, jurisdiction, structure information, highway IDs, and functional class.
- Managed and processed **~200 GB/year** of ARAN pavement survey data, transferring field-collected LCMS imagery and GPS datasets to Azure Blob Storage and validating and segmenting continuous survey data in Roadware Vision for AMS use.
- Developed a clustering-based methodology to select 7 representative pavement sections from ~1,800 MTO Asset Management System (AMS) records using IRI, DMI, PCI, and rutting; applied K-Means with elbow-method cluster selection and used PCA and radar charts for visualization, with **5 of 7** selected sections matching vendor recommendations.

Research Student

May 2024 – Apr 2025

Michael Garron Hospital, East York, ON

- Contributed to an AHRQ-funded Diagnostic Center of Excellence systematic review, screening records from a pool of 10,000+ citations in Covidence against predefined eligibility criteria and conducting full-text review of approximately 20 studies.

SELECTED PROJECTS

Topic Modelling of Patient Free-Text Surveys (LDA) Python, NLTK, Gensim, Hyperopt, guidedlda

- Built an LDA topic-modeling pipeline to extract themes from patient free-text surveys; processed responses at **< 0.05 s/record**; integrated preprocessing (cleaning, tokenization, lemmatization).
- Tuned **10** hyperparameters with TPE (Bayesian optimization) to improve coherence/perplexity; delivered interpretable topic summaries for stakeholders.
- Presented this work at Diagnostic Excellence 2024 at the University of Minnesota and the 2024 Inter-University Workshop at the State University of New York.

ACHIEVEMENTS

UTMIST Hackathon — Winner University of Toronto

Mar 2023

Top team for an applied ML/NLP challenge; rapid prototyping and model evaluation under time constraints.

PUBLICATIONS

Reimaging Health Equity, Equality, and Justice: A Fresh Vision

Longwoods Insights (2024).

McCallum, E; Scala, N; Mason, T; **Zheng, H**; Priore, A; Coppinger, T; Sochaniwskyj, M; Pettit A; Hill, MA; Shearkhani, S.

longwoods.com/content/27405

Opportunities and Challenges for the Application of AI in Evaluations of Healthcare Services Longwoods Insights (2025).

Mateus, L; Qu, Y; **Zheng, H**; Shearkhani, S

longwoods.com/content/27544